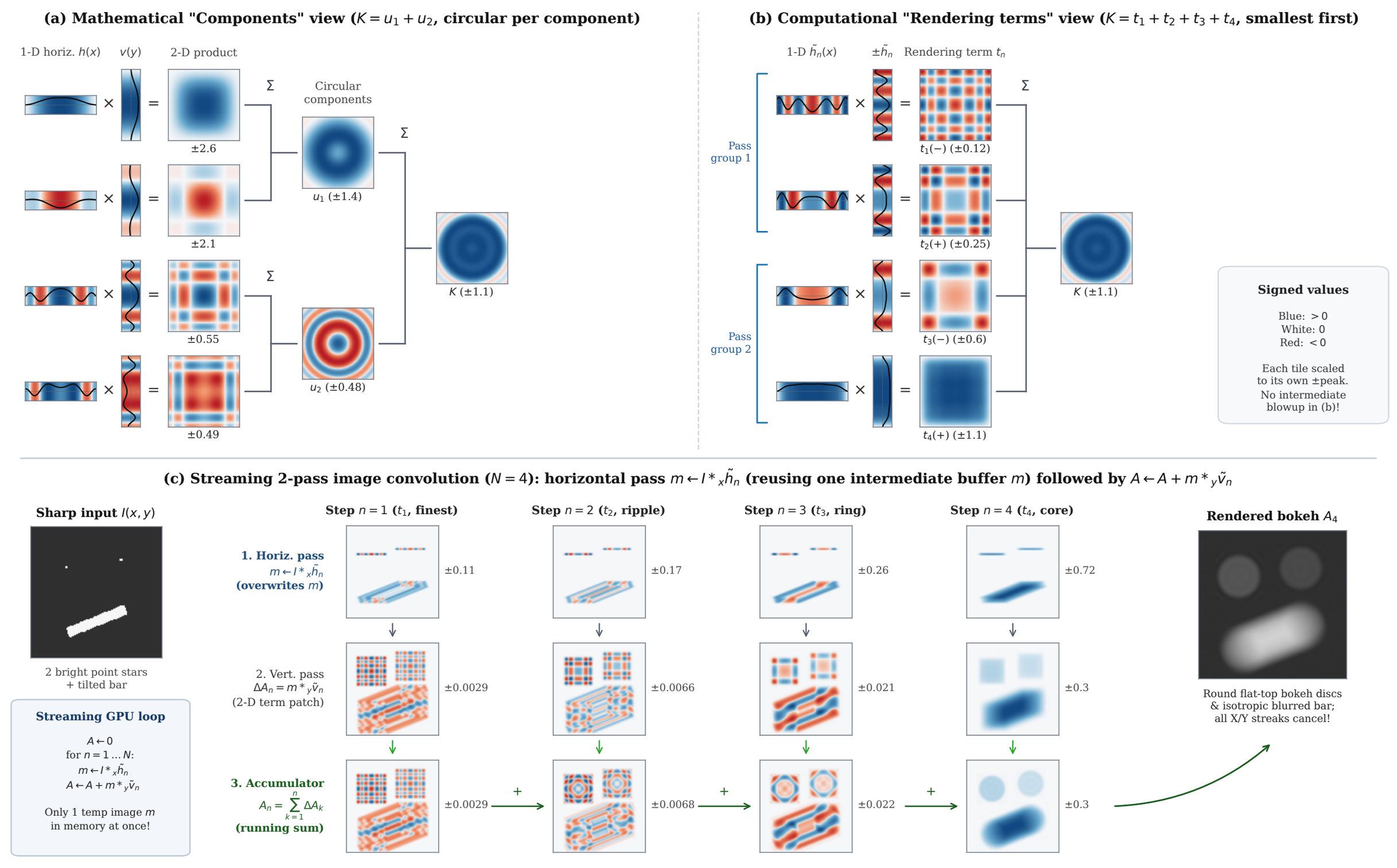

(a) Mathematical "Components" view ($K = u_1 + u_2$, circular per component)
1-D horiz. $h(x)$
$v(y)$
2-D product
±2.6
±2.1
±0.55
±0.49
Σ
Circular components
$u_1$ (±1.4)
$u_2$ (±0.48)
$K$ (±1.1)
(b) Computational "Rendering terms" view ($K = t_1 + t_2 + t_3 + t_4$, smallest first)
1-D $\tilde{h}_n(x)$
$\pm\tilde{h}_n$
Rendering term $t_n$
Pass group 1
Pass group 2
$t_1(-)$ (±0.12)
$t_2(+)$ (±0.25)
$t_3(-)$ (±0.6)
$t_4(+)$ (±1.1)
$K$ (±1.1)
Signed values
Blue: > 0
White: 0
Red: < 0
Each tile scaled to its own ±peak.
No intermediate blowup in (b)!
(c) Streaming 2-pass image convolution ($N = 4$): horizontal pass $m \leftarrow I *_x \tilde{h}_n$ (reusing one intermediate buffer $m$) followed by $A \leftarrow A + m *_y \tilde{v}_n$
Sharp input $I(x, y)$
2 bright point stars + tilted bar
Streaming GPU loop
$A \leftarrow 0$
for $n = 1 \dots N$:
$m \leftarrow I *_x \tilde{h}_n$
$A \leftarrow A + m *_y \tilde{v}_n$
Only 1 temp image $m$ in memory at once!
Step $n = 1$ ($t_1$, finest)
Step $n = 2$ ($t_2$, ripple)
Step $n = 3$ ($t_3$, ring)
Step $n = 4$ ($t_4$, core)
1. Horiz. pass $m \leftarrow I *_x \tilde{h}_n$ (overwrites $m$)
2. Vert. pass $\Delta A_n = m *_y \tilde{v}_n$ (2-D term patch)
3. Accumulator $A_n = \sum_{k=1}^{n} \Delta A_k$ (running sum)
±0.11
±0.17
±0.26
±0.72
±0.0029
±0.0066
±0.021
±0.3
±0.0029
±0.0068
±0.022
±0.3
+
Rendered bokeh $A_4$
Round flat-top bokeh discs & isotropic blurred bar; all X/Y streaks cancel!